Exercise 15.7.49

Mass, Center of Mass, and Moment of Inertia: Exact Values

Problem

Let E be the solid in the first octant bounded by the cylinder $x^2 + y^2 = 1$ and the planes $y = z$, $x = 0$, and $z = 0$ with the density function $\rho(x, y, z) = 1 + x + y + z$. Use a computer algebra system to find the exact values of the following quantities for E.

(a) The mass

(b) The center of mass

(c) The moment of inertia about the z-axis

The solid, the coordinates, and the notation

The symbol E denotes the solid region. The symbol ρ denotes the density, the mass per unit volume. The symbol dV denotes the volume element. The symbol m denotes the mass, $(\bar{x}, \bar{y}, \bar{z})$ denotes the center of mass, and I_z denotes the moment of inertia about the z-axis.

The first octant is the set $x \geq 0$, $y \geq 0$, $z \geq 0$. The four surfaces are:

- the cylinder $x^2 + y^2 = 1$, a vertical round wall;
- the plane $y = z$, a slanted roof that rises in the y-direction;
- the plane $x = 0$, a flat back wall;
- the plane $z = 0$, the floor.

For a fixed point (x, y) in the quarter disk $x^2 + y^2 \leq 1$, a vertical line enters E at $z = 0$ and leaves E at $z = y$. Thus E is z-simple over the quarter disk. The solid is a wedge with a curved wall.

Cylindrical coordinates. The wall is a circle, and the density contains $x^2 + y^2$. Therefore cylindrical coordinates are natural. The symbols are:

- r is the distance from the z-axis, so $r \geq 0$;
- θ is the angle from the positive x-axis in the xy-plane;
- z is the same height as in rectangular coordinates.

The conversion is $x = r\cos\theta$, $y = r\sin\theta$, $z = z$, and the volume element is

$$dV = r\,dz\,dr\,d\theta.$$

The factor r is the Jacobian of the change of variables; it is not optional. In cylindrical coordinates the solid is

$$E = \{(r,\theta,z) : 0 \le \theta \le \tfrac{\pi}{2},\ 0 \le r \le 1,\ 0 \le z \le r\sin\theta\},$$

and the density is

$$\rho = 1 + r\cos\theta + r\sin\theta + z.$$

Figure 1 shows E, and Figure 2 shows the base and the face $x = 0$.

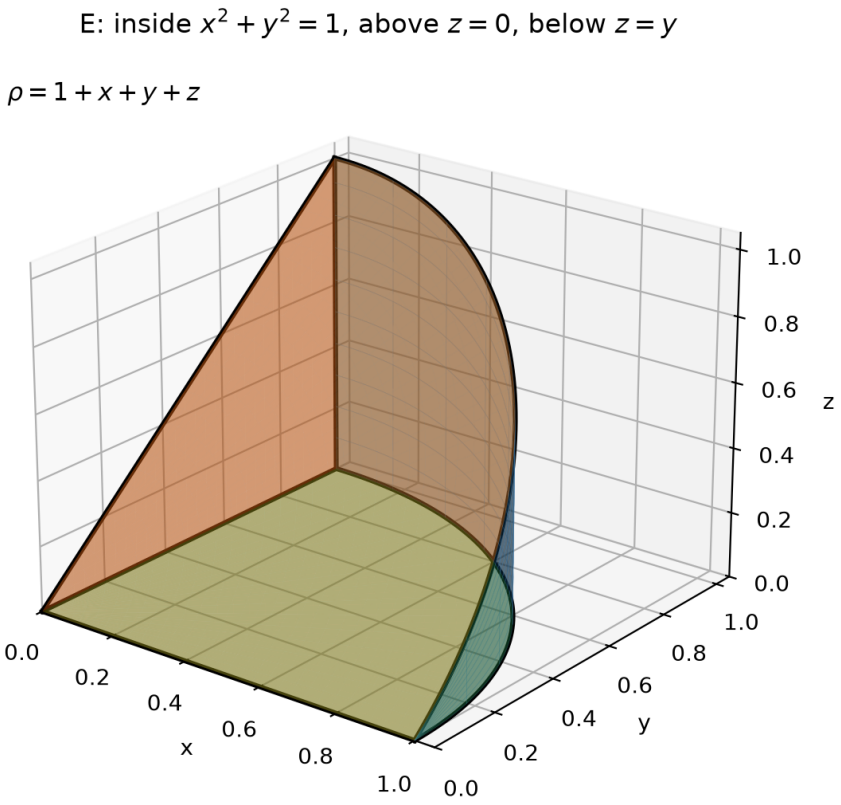


Figure 1: The solid E of Exercise 15.7.49. The blue sheet is the cylinder $x^2 + y^2 = 1$. The orange sheet is the slanted roof $z = y$ over the quarter disk. The green sheet is the floor $z = 0$. The red sheet is the back wall $x = 0$. The density is $\rho = 1 + x + y + z$.

Part (a): the mass

The mass is the integral of the density over the solid:

$$m = \iiint_E \rho\,dV = \int_0^{\pi/2}\int_0^1\int_0^{r\sin\theta} (1 + r\cos\theta + r\sin\theta + z)\,r\,dz\,dr\,d\theta.$$

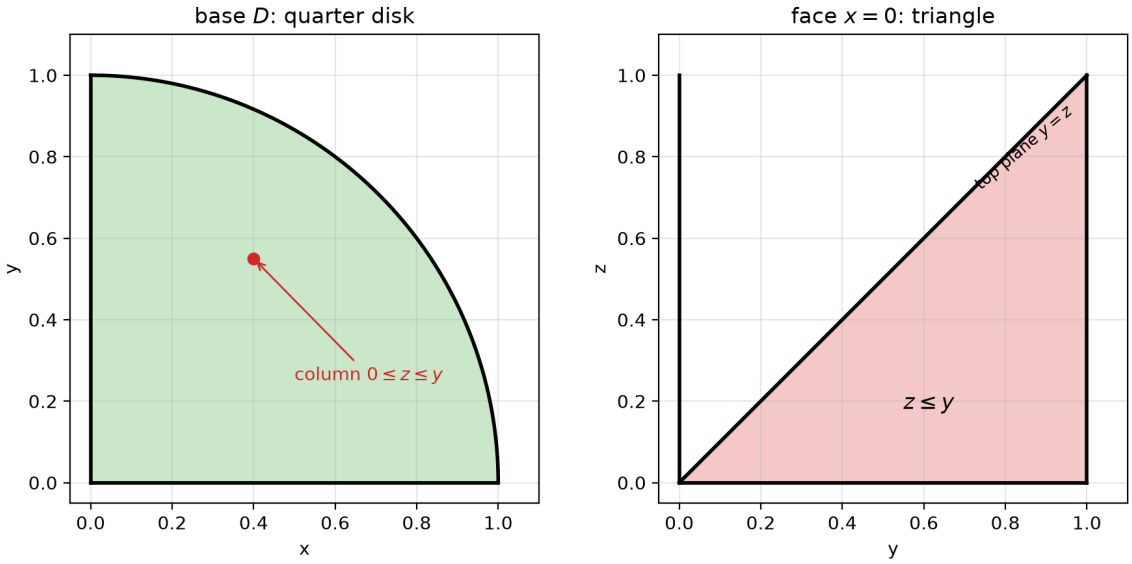


Figure 2: Left: the base D is the quarter disk $x^2 + y^2 \le 1$ in the first quadrant; each point carries the column $0 \le z \le y$. Right: the face $x = 0$ is the triangle $0 \le z \le y \le 1$, which shows why the roof is $z = y$.

Write $c = \cos\theta$ and $s = \sin\theta$ to keep the display short. First integrate with respect to z:

$$\int_0^{rs} (1 + rc + rs + z)\, dz = (1 + rc + rs)rs + \frac{(rs)^2}{2} = rs + r^2 sc + \frac{3}{2} r^2 s^2.$$

Multiplication by the Jacobian r gives

$$r^2 s + r^3 sc + \frac{3}{2} r^3 s^2.$$

Now integrate with respect to r from 0 to 1:

$$\int_0^1 \left(r^2 s + r^3 sc + \frac{3}{2} r^3 s^2 \right) dr = \frac{s}{3} + \frac{sc}{4} + \frac{3s^2}{8}.$$

Finally integrate with respect to θ from 0 to $\pi/2$. The three standard values are

$$\int_0^{\pi/2} \sin\theta\, d\theta = 1, \qquad \int_0^{\pi/2} \sin\theta\cos\theta\, d\theta = \frac{1}{2}, \qquad \int_0^{\pi/2} \sin^2\theta\, d\theta = \frac{\pi}{4}.$$

Therefore

$$m = \frac{1}{3} + \frac{1}{8} + \frac{3}{8} \cdot \frac{\pi}{4} = \frac{11}{24} + \frac{3\pi}{32} = \frac{44 + 9\pi}{96} \approx 0.752858.$$

$$\boxed{m = \frac{44 + 9\pi}{96} \approx 0.752858}$$

Part (b): the center of mass

The coordinates of the center of mass are the first moments divided by the mass:

$$\bar{x} = \frac{M_{yz}}{m}, \qquad \bar{y} = \frac{M_{xz}}{m}, \qquad \bar{z} = \frac{M_{xy}}{m},$$

where

$$M_{yz} = \iiint_E x\rho\,dV, \qquad M_{xz} = \iiint_E y\rho\,dV, \qquad M_{xy} = \iiint_E z\rho\,dV.$$

The subscripts name the coordinate plane with respect to which the moment is taken. For example, M_{yz} is the moment with respect to the plane $x = 0$.

The moment M_{yz}. Here $x\rho\,dV = (r\cos\theta)\rho\, r\,dz\,dr\,d\theta = r^2 c\rho\,dz\,dr\,d\theta$:

$$M_{yz} = \int_0^{\pi/2}\int_0^1\int_0^{rs} r^2 c\,(1 + rc + rs + z)\,dz\,dr\,d\theta.$$

The inner integral is

$$r^2 c\Big[(1 + rc + rs)rs + \frac{r^2 s^2}{2}\Big] = r^3 sc + r^4 sc^2 + \frac{3}{2}r^4 s^2 c.$$

Then

$$\int_0^1 \Big(r^3 sc + r^4 sc^2 + \frac{3}{2}r^4 s^2 c\Big)\,dr = \frac{sc}{4} + \frac{sc^2}{5} + \frac{3s^2 c}{10},$$

and

$$M_{yz} = \frac{1}{8} + \frac{1}{15} + \frac{1}{10} = \frac{7}{24}.$$

The three elementary integrals used here are $\int_0^{\pi/2} sc\,d\theta = 1/2$, $\int_0^{\pi/2} sc^2\,d\theta = 1/3$, and $\int_0^{\pi/2} s^2 c\,d\theta = 1/3$. Hence

$$\bar{x} = \frac{7/24}{(44 + 9\pi)/96} = \frac{28}{44 + 9\pi} \approx 0.387413.$$

The moment M_{xz}. Here $y\rho\,dV = r^2 s\rho\,dz\,dr\,d\theta$, and the same steps give

$$M_{xz} = \frac{\pi}{16} + \frac{4}{15},$$

because the θ-integrals are $\int_0^{\pi/2} s^2\,d\theta = \pi/4$, $\int_0^{\pi/2} s^2 c\,d\theta = 1/3$, and $\int_0^{\pi/2} s^3\,d\theta = 2/3$. Therefore

$$\bar{y} = \frac{\pi/16 + 4/15}{(44 + 9\pi)/96} = \frac{2(15\pi + 64)}{5(44 + 9\pi)} \approx 0.615012.$$

The moment M_{xy}. Here $z\rho\,dV = z\rho\, r\,dz\,dr\,d\theta$, and

$$\int_0^{rs} z(1 + rc + rs + z)\,dz = \frac{(1 + rc + rs)r^2 s^2}{2} + \frac{r^3 s^3}{3} = \frac{r^2 s^2}{2} + \frac{r^3 s^2 c}{2} + \frac{5r^3 s^3}{6}.$$

Multiplication by r and integration give

$$M_{xy} = \frac{1}{8}\cdot\frac{\pi}{4} + \frac{1}{10}\cdot\frac{1}{3} + \frac{1}{6}\cdot\frac{2}{3} = \frac{\pi}{32} + \frac{13}{90},$$

so

$$\bar{z} = \frac{\pi/32 + 13/90}{(44 + 9\pi)/96} = \frac{45\pi + 208}{15(44 + 9\pi)} \approx 0.322264.$$

$$\boxed{(\bar{x}, \bar{y}, \bar{z}) = \Big(\frac{28}{44 + 9\pi},\ \frac{2(15\pi + 64)}{5(44 + 9\pi)},\ \frac{45\pi + 208}{15(44 + 9\pi)}\Big) \approx (0.387413,\ 0.615012,\ 0.322264)}$$

Part (c): the moment of inertia about the z-axis

The distance of a point from the z-axis is r, so the squared distance is r^2 and

$$I_z = \iiint_E r^2 \rho \, dV = \int_0^{\pi/2} \int_0^1 \int_0^{rs} r^3 (1 + rc + rs + z) \, dz \, dr \, d\theta.$$

The inner integral is

$$r^3 \Big[(1 + rc + rs) rs + \frac{r^2 s^2}{2} \Big] = r^4 s + r^5 sc + \frac{3}{2} r^5 s^2.$$

Then

$$\int_0^1 \Big(r^4 s + r^5 sc + \frac{3}{2} r^5 s^2 \Big) \, dr = \frac{s}{5} + \frac{sc}{6} + \frac{s^2}{4},$$

and

$$I_z = \frac{1}{5} + \frac{1}{6} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{\pi}{4} = \frac{17}{60} + \frac{\pi}{16} \approx 0.479683.$$

$$\boxed{I_z = \frac{\pi}{16} + \frac{17}{60} \approx 0.479683}$$

Verification

The results were computed twice: once exactly with the computer algebra system SymPy, and once numerically with a Gauss–Legendre product rule and with a Monte Carlo average on E. Table 1 lists the values.

Quantity	Exact value	Exact (decimal)	Numeric
m	$(44 + 9\pi)/96$	0.752857645	0.752857645
$\bar{x}$	$28/(44 + 9\pi)$	0.387412772	0.387412772
$\bar{y}$	$2(15\pi + 64)/(5(44 + 9\pi))$	0.615011630	0.615011630
$\bar{z}$	$(45\pi + 208)/(15(44 + 9\pi))$	0.322264397	0.322264397
I_z	$\pi/16 + 17/60$	0.479682874	0.479682874

Table 1: Exact and numerical values for Exercise 15.7.49. The numerical column comes from a Gauss–Legendre rule on E; a Monte Carlo average agrees within its standard error.

A check with unit density. If $\rho = 1$, the integrals in this document give

$$\int_0^{\pi/2} \int_0^1 \int_0^{rs} r \, dz \, dr \, d\theta = \int_0^{\pi/2} \int_0^1 r^2 s \, dr \, d\theta = \frac{1}{3}.$$

This is the volume of E, and it is the volume requested in Exercise 15.7.23, which uses the same solid. The value $1/3$ is consistent with Figure 2: the base D is a quarter disk of area $\pi/4$, and the average height of the roof over D is

$$\frac{1}{\pi/4} \iint_D y \, dA = \frac{4}{\pi} \cdot \frac{1}{3} = \frac{4}{3\pi},$$

so the volume is $\frac{\pi}{4} \cdot \frac{4}{3\pi} = \frac{1}{3}$. The results for m, $\bar{x}$, $\bar{y}$, $\bar{z}$, and I_z pass the checks in `part1_step1.py`, `part2_step1.py`, and `part3_step1.py`; each script prints `PASS`.

How to solve problems of this kind

Every problem that asks for the mass, the center of mass, or a moment of inertia of a solid of given density follows the same pattern: write one master integral for each requested quantity, evaluate the integrals in the coordinate system that matches the bounding surfaces, and test the answer against elementary bounds. The steps below apply to this exercise and to any other exercise of the same type.

1. **Write the solid as a set of inequalities before you integrate.** Here the first octant, the cylinder $x^2+y^2=1$, and the planes $y=z$ and $z=0$ give $E=\{0\le y\le 1,\ 0\le z\le y,\ 0\le x\le \sqrt{1-y^2}\}$, and the cylindrical form is $0\le\theta\le\pi/2$, $0\le r\le 1$, $0\le z\le r\sin\theta$. The density is $\rho(x,y,z)=1+x+y+z$, which is the mass per unit volume. The total mass m is the integral of ρ over E.

2. **Choose the coordinates from the surfaces, not from the density.** A circular cylinder about the z-axis, as here, becomes $r=$ constant in cylindrical coordinates (r,θ,z), so the outer two limits become constants. Use spherical coordinates (ϱ,θ,φ) instead when the surfaces are spheres or cones, and use rectangular coordinates when all six faces are planes.

3. **Write the four master integrals once.** With dV the volume element,

$$m=\iiint_E \rho\,dV,\qquad \bar{x}=\frac{M_{yz}}{m},\quad \bar{y}=\frac{M_{xz}}{m},\quad \bar{z}=\frac{M_{xy}}{m},\qquad I_z=\iiint_E (x^2+y^2)\,\rho\,dV.$$

   The first moments are $M_{yz}=\iiint x\rho\,dV$, $M_{xz}=\iiint y\rho\,dV$, and $M_{xy}=\iiint z\rho\,dV$. The symbol M_{yz} is the moment of the solid about the plane $x=0$, and the factor x is the distance from that plane. The center of mass is $(\bar{x},\bar{y},\bar{z})$, and I_z is the moment of inertia about the z-axis. In I_z the factor x^2+y^2 is the *squared* distance from the point (x,y,z) to the z-axis, because the distance itself is $\sqrt{x^2+y^2}$. Never use x^2 alone for I_z.

4. **Expand the density and use symmetry.** A polynomial density multiplies into monomials, and each monomial integrates separately. If E is symmetric in a coordinate and the integrand is odd in that coordinate, the integral is zero. In this exercise E has no symmetry plane in x or y, so all three coordinates of the center of mass are nonzero; the value $\bar{x}<\bar{y}$ is a consequence of the roof $z=y$, which makes the solid taller at large y.

5. **Get the exact value, then the decimal.** Ask the computer algebra system for the closed form, for example $(44+9\pi)/96$, and compute the decimal separately. A decimal alone cannot show a small algebra error, but an exact value that equals a simple combination of π and rational numbers is strong evidence of a correct set-up.

6. **Test the answer against bounds that need no integration.** The mass satisfies $\rho_{\min}V(E)\le m\le\rho_{\max}V(E)$, so here $\frac{1}{3}\le m\le 4\cdot\frac{1}{3}$. The moment of inertia satisfies $m\,r_{\min}^2\le I_z\le m\,r_{\max}^2$, where $r_{\min}$ and $r_{\max}$ are the smallest and largest distances from the z-axis in E; here $0\le I_z\le m$. The center of mass must lie inside E. Finally, replace the density by 1: the same integrals must return the volume of E, which is $\frac{1}{3}$ here.

7. **Repeat the arithmetic a second way.** Use a product Gauss–Legendre rule or a Monte Carlo average, and compare with the exact value. When both agree to every digit that the rule can support, the set-up and the arithmetic are almost certainly correct.